and the scalarization method for the (unconstrained) multicriterion problem

$$\text{minimize (w.r.t. } \mathbf{R}_+^{m+1}) \quad F(x) = (f_1(x), \ldots, f_m(x), f_0(x)) \tag{5.44}$$

(see §4.7.4). In scalarization, we choose a positive vector $\tilde{\lambda}$, and minimize the scalar function $\tilde{\lambda}^T F(x)$; any minimizer is guaranteed to be Pareto optimal. Since we can scale $\tilde{\lambda}$ by a positive constant, without affecting the minimizers, we can, without loss of generality, take $\tilde{\lambda} = (\lambda, 1)$. Thus, in scalarization we minimize the function

$$\tilde{\lambda}^T F(x) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x),$$

which is exactly the Lagrangian for the problem (5.43).

To establish that every Pareto optimal point of a convex multicriterion problem minimizes the function $\tilde{\lambda}^T F(x)$ for some nonnegative weight vector $\tilde{\lambda}$, we considered the set $\mathcal{A}$, defined in (4.62),

$$\mathcal{A} = \{t \in \mathbf{R}^{m+1} \mid \exists x \in \mathcal{D},\ f_i(x) \leq t_i,\ i = 0, \ldots, m\},$$

which is exactly the same as the set $\mathcal{A}$ defined in (5.37), that arises in Lagrange duality. Here too we constructed the required weight vector as a supporting hyperplane to the set, at an arbitrary Pareto optimal point. In multicriterion optimization, we interpret the components of the weight vector as giving the relative weights between the objective functions. When we fix the last component of the weight vector (associated with f_0) to be one, the other weights have the interpretation of the cost relative to f_0, *i.e.*, the cost relative to the objective.

5.4 Saddle-point interpretation

In this section we give several interpretations of Lagrange duality. The material of this section will not be used in the sequel.

5.4.1 Max-min characterization of weak and strong duality

It is possible to express the primal and the dual optimization problems in a form that is more symmetric. To simplify the discussion we assume there are no equality constraints; the results are easily extended to cover them.

First note that

$$\begin{aligned} \sup_{\lambda \succeq 0} L(x, \lambda) &= \sup_{\lambda \succeq 0} \left(f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) \right) \\ &= \begin{cases} f_0(x) & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ \infty & \text{otherwise.} \end{cases} \end{aligned}$$

Indeed, suppose x is not feasible, and $f_i(x) > 0$ for some i. Then $\sup_{\lambda \succeq 0} L(x, \lambda) = \infty$, as can be seen by choosing $\lambda_j = 0$, $j \neq i$, and $\lambda_i \to \infty$. On the other hand, if $f_i(x) \leq 0$, $i = 1, \ldots, m$, then the optimal choice of λ is $\lambda = 0$ and $\sup_{\lambda \succeq 0} L(x, \lambda) = f_0(x)$. This means that we can express the optimal value of the primal problem as

$$p^\star = \inf_x \sup_{\lambda \succeq 0} L(x, \lambda).$$

By the definition of the dual function, we also have

$$d^\star = \sup_{\lambda \succeq 0} \inf_x L(x, \lambda).$$

Thus, weak duality can be expressed as the inequality

$$\sup_{\lambda \succeq 0} \inf_x L(x, \lambda) \leq \inf_x \sup_{\lambda \succeq 0} L(x, \lambda), \tag{5.45}$$

and strong duality as the equality

$$\sup_{\lambda \succeq 0} \inf_x L(x, \lambda) = \inf_x \sup_{\lambda \succeq 0} L(x, \lambda).$$

Strong duality means that the order of the minimization over x and the maximization over $\lambda \succeq 0$ can be switched without affecting the result.

In fact, the inequality (5.45) does not depend on any properties of L: We have

$$\sup_{z \in Z} \inf_{w \in W} f(w, z) \leq \inf_{w \in W} \sup_{z \in Z} f(w, z) \tag{5.46}$$

for any $f : \mathbf{R}^n \times \mathbf{R}^m \to \mathbf{R}$ (and any $W \subseteq \mathbf{R}^n$ and $Z \subseteq \mathbf{R}^m$). This general inequality is called the *max-min inequality*. When equality holds, *i.e.*,

$$\sup_{z \in Z} \inf_{w \in W} f(w, z) = \inf_{w \in W} \sup_{z \in Z} f(w, z) \tag{5.47}$$

we say that f (and W and Z) satisfy the *strong max-min property* or the *saddle-point* property. Of course the strong max-min property holds only in special cases, for example, when $f : \mathbf{R}^n \times \mathbf{R}^m \to \mathbf{R}$ is the Lagrangian of a problem for which strong duality obtains, $W = \mathbf{R}^n$, and $Z = \mathbf{R}^m_+$.

5.4.2 Saddle-point interpretation

We refer to a pair $\tilde{w} \in W$, $\tilde{z} \in Z$ as a *saddle-point* for f (and W and Z) if

$$f(\tilde{w}, z) \leq f(\tilde{w}, \tilde{z}) \leq f(w, \tilde{z})$$

for all $w \in W$ and $z \in Z$. In other words, $\tilde{w}$ minimizes $f(w, \tilde{z})$ (over $w \in W$) and $\tilde{z}$ maximizes $f(\tilde{w}, z)$ (over $z \in Z$):

$$f(\tilde{w}, \tilde{z}) = \inf_{w \in W} f(w, \tilde{z}), \qquad f(\tilde{w}, \tilde{z}) = \sup_{z \in Z} f(\tilde{w}, z).$$

This implies that the strong max-min property (5.47) holds, and that the common value is $f(\tilde{w}, \tilde{z})$.

Returning to our discussion of Lagrange duality, we see that if $x^\star$ and $\lambda^\star$ are primal and dual optimal points for a problem in which strong duality obtains, they form a saddle-point for the Lagrangian. The converse is also true: If (x, λ) is a saddle-point of the Lagrangian, then x is primal optimal, λ is dual optimal, and the optimal duality gap is zero.

5.4.3 Game interpretation

We can interpret the max-min inequality (5.46), the max-min equality (5.47), and the saddle-point property, in terms of a continuous *zero-sum game*. If the first player chooses $w \in W$, and the second player selects $z \in Z$, then player 1 pays an amount $f(w, z)$ to player 2. Player 1 therefore wants to minimize f, while player 2 wants to maximize f. (The game is called continuous since the choices are vectors, and not discrete.)

Suppose that player 1 makes his choice first, and then player 2, after learning the choice of player 1, makes her selection. Player 2 wants to maximize the payoff $f(w, z)$, and so will choose $z \in Z$ to maximize $f(w, z)$. The resulting payoff will be $\sup_{z \in Z} f(w, z)$, which depends on w, the choice of the first player. (We assume here that the supremum is achieved; if not the optimal payoff can be arbitrarily close to $\sup_{z \in Z} f(w, z)$.) Player 1 knows (or assumes) that player 2 will follow this strategy, and so will choose $w \in W$ to make this worst-case payoff to player 2 as small as possible. Thus player 1 chooses

$$\operatorname*{argmin}_{w \in W} \sup_{z \in Z} f(w, z),$$

which results in the payoff

$$\inf_{w \in W} \sup_{z \in Z} f(w, z)$$

from player 1 to player 2.

Now suppose the order of play is reversed: Player 2 must choose $z \in Z$ first, and then player 1 chooses $w \in W$ (with knowledge of z). Following a similar argument, if the players follow the optimal strategy, player 2 should choose $z \in Z$ to maximize $\inf_{w \in W} f(w, z)$, which results in the payoff of

$$\sup_{z \in Z} \inf_{w \in W} f(w, z)$$

from player 1 to player 2.

The max-min inequality (5.46) states the (intuitively obvious) fact that it is better for a player to go second, or more precisely, for a player to know his or her opponent's choice before choosing. In other words, the payoff to player 2 will be larger if player 1 must choose first. When the saddle-point property (5.47) holds, there is no advantage to playing second.

If $(\tilde{w}, \tilde{z})$ is a saddle-point for f (and W and Z), then it is called a *solution* of the game; $\tilde{w}$ is called the optimal choice or strategy for player 1, and $\tilde{z}$ is called

the optimal choice or strategy for player 2. In this case there is no advantage to playing second.

Now consider the special case where the payoff function is the Lagrangian, $W = \mathbf{R}^n$ and $Z = \mathbf{R}^m_+$. Here player 1 chooses the primal variable x, while player 2 chooses the dual variable $\lambda \succeq 0$. By the argument above, the optimal choice for player 2, if she must choose first, is any $\lambda^\star$ which is dual optimal, which results in a payoff to player 2 of $d^\star$. Conversely, if player 1 must choose first, his optimal choice is any primal optimal $x^\star$, which results in a payoff of $p^\star$.

The optimal duality gap for the problem is exactly equal to the advantage afforded the player who goes second, *i.e.*, the player who has the advantage of knowing his or her opponent's choice before choosing. If strong duality holds, then there is no advantage to the players of knowing their opponent's choice.

5.4.4 Price or tax interpretation

Lagrange duality has an interesting economic interpretation. Suppose the variable x denotes how an enterprise operates and $f_0(x)$ denotes the cost of operating at x, *i.e.*, $-f_0(x)$ is the profit (say, in dollars) made at the operating condition x. Each constraint $f_i(x) \leq 0$ represents some limit, such as a limit on resources (*e.g.*, warehouse space, labor) or a regulatory limit (*e.g.*, environmental). The operating condition that maximizes profit while respecting the limits can be found by solving the problem

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m. \end{array}$$

The resulting optimal profit is $-p^\star$.

Now imagine a second scenario in which the limits can be violated, by paying an additional cost which is linear in the amount of violation, measured by f_i. Thus the payment made by the enterprise for the ith limit or constraint is $\lambda_i f_i(x)$. Payments are also made *to* the firm for constraints that are not tight; if $f_i(x) < 0$, then $\lambda_i f_i(x)$ represents a payment to the firm. The coefficient λ_i has the interpretation of the price for violating $f_i(x) \leq 0$; its units are dollars per unit violation (as measured by f_i). For the same price the enterprise can sell any 'unused' portion of the ith constraint. We assume $\lambda_i \geq 0$, *i.e.*, the firm must pay for violations (and receives income if a constraint is not tight).

As an example, suppose the first constraint in the original problem, $f_1(x) \leq 0$, represents a limit on warehouse space (say, in square meters). In this new arrangement, we open the possibility that the firm can rent extra warehouse space at a cost of λ_1 dollars per square meter and also rent out unused space, at the same rate.

The total cost to the firm, for operating condition x, and constraint prices λ_i, is $L(x, \lambda) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x)$. The firm will obviously operate so as to minimize its total cost $L(x, \lambda)$, which yields a cost $g(\lambda)$. The dual function therefore represents the optimal cost to the firm, as a function of the constraint price vector λ. The optimal dual value, $d^\star$, is the optimal cost to the enterprise under the least favorable set of prices.